

JEE MAIN REVISION MASTER: PRACTICAL ORGANIC CHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Qualitative & Quantitative Organic Analysis	3 PYQs	Practical Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Qualitative & Quantitative Organic Analysis	$\%N = \frac{1.4 \times N \times V}{w} \times 100$ $\text{Prussian Blue: } \text{Fe}_4[\text{Fe}(\text{CN})_6]_3$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Qualitative & Quantitative Organic Analysis (Moderate Yield — 3 PYQs)

Core Principle: Lassaigne's test for nitrogen gives Prussian blue precipitate of ferric ferrocyanide $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$. If both nitrogen and sulphur are present, sodium fusion gives blood-red $\text{Fe}(\text{SCN})^2+$. Kjeldahl's method for nitrogen: $\%N = (1.4 \times N \times V / w)$. Carius method for halogens uses concentrated $\text{HNO}_3 + \text{AgNO}_3$ to precipitate AgX .

■ **Caution / JEE Pitfall:** Kjeldahl's method is NOT applicable to compounds containing nitrogen in nitro ($-\text{NO}_2$), azo ($-\text{N}=\text{N}-$), or ring nitrogen (pyridine) because they do not form ammonium sulphate.

■ **Speed Trick:** Lassaigne's test fails for hydrazine (NH_2NH_2) because it contains no carbon to form cyanide ion (CN^-).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Statement-I: Nitrogen in pyridine cannot be estimated by Kjeldahl's method. Statement-II: Nitrogen in pyridine changes to ammonium sulphate ..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Match List-I with List-II and select the correct options List-I (Functional group) List-II (Functional group reagent) (a) (i) Neutral ferric..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "In the preparation of sodium fusion extract, the purpose of fusing organic compound with a piece of sodium metal is to..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Statement-I: Nitrogen in pyridine cannot be estimated by Kjeldahl's method. Statement-II: Nitrogen in pyridine changes to ammonium sulphate when heated with conc. H_2SO_4 in Kjeldahl's method. Read the above given statements and choose the correct answer from the given options.

- (A) Statement I is true but Statement II is false
- (B) Both Statement I and Statement II are false
- (C) Both Statement I and Statement II are true
- (D) Statement I is false but Statement II is true

Correct Answer: None

Solution Analysis: Statement-I is true. In Kjeldahl's method, the nitrogen present in an organic compound is converted into ammonium sulphate, $(\text{NH}_4)_2\text{SO}_4$, by heating with concentrated H_2SO_4 . But this method does not work for all nitrogen-containing compounds. It fails for compounds in which nitrogen is present in certain special forms, such as: nitro compounds, azo compounds, diazo compounds, heterocyclic compounds like pyridine. In pyridine, nitrogen is part of the aromatic ring. This nitrogen does not get converted into ammonium sulphate on heating with concentrated